

Reduction of fan noise emission by enclosure modification

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In the present study, a structural modification approach to the reduction of sound power radiated by an axial fan mounted to an enclosure is described. An axial fan operating in free-space exhibits a dipole-like radiation pattern at low frequencies when the source is acoustically compact. When a fan is mounted to an enclosure, however, the sound radiation pattern becomes monopole-like since the interaction between the sound fields on the two sides of the fan is prevented by the enclosure: thus the fan can radiate sound more efficiently than when operating in free-space. In addition, it is possible for the source level to be amplified by coupling with interior resonances of the enclosure. The radiation enhancement caused by both effects can be suppressed by introducing an acoustical path that allows the sound fields on both sides of the fan to interact with each other while also changing the resonance frequencies of, and providing damping, for the acoustical cavity, thus reducing the radiation efficiency of the fan towards its free-space value. The effect of the latter type of modification is demonstrated here by using a consumer electronics device equipped with two fans. The sound fields corresponding to various enclosure modifications were visualized by using near-field acoustical holography, and the modifications' effects are quantified in terms of the radiated sound power.

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1 INTRODUCTION

Axial fans are widely used for electronic cooling, but their use often results in a noise nuisance. To find efficient noise control solutions for these cases, it is important to understand the noise generation mechanisms of a fan. Among the various source mechanisms

that contribute to the overall noise emission, the most dominant source type, especially in the case of the low-speed fans that are considered in the present study, is known to be the dipole-like source that is generated by the fluctuating pressure distribution on the surface of the blades. Thus, an axial fan operating in free-space can be modeled as a point dipole whose source strength is proportional to the magnitude of a fluctuating point force so long as the compact source assumption holds true (i.e., when the dimensions of the fan are much smaller than an acoustic wavelength). It is also known that the source strength of an aerodynamically-induced dipole of the latter kind is proportional to the sixth power of the flow speed¹. Therefore, the simplest remedy for the latter type of noise is to decrease the rotational speed of a fan while maintaining the volume flow rate by increasing the size of a fan in order to achieve the same cooling performance; the application of the latter approach is often restricted, however, either by the design layout or cost issues.

When a fan is mounted to an enclosure, the sound radiation pattern becomes monopole-like rather than dipole-like because the interaction between the sound fields on both sides of the fan is prevented by the encl-

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sure. Since a monopole is a more efficient radiator than a dipole², the fan mounted to the enclosure results in a higher radiated sound power level than the same fan operating in free-space. Secondly, the fan's radiated sound power level can be amplified by the effect of the enclosure's interior acoustical resonances; when the fan's blade passing frequency coincides with those resonance frequencies, for example.

The structural modification scheme described in the present study is based on a physical understanding of the latter aspects of acoustical sources. The main idea is to introduce an acoustical path between the sound fields on the two sides of the fan so that cancellation between the sound fields that are out-of-phase with each other can occur, thus reducing the overall sound radiation, while at the same time weakening the effect of the interior acoustical resonances.

In the present study, the effect of an enclosure as a structure that hinders the interaction between two sound fields is demonstrated by using simulation results, while the effect of the acoustical cavity resonance is demonstrated experimentally by using a model enclosure. Finally, experimental results obtained by the application of these principles to a consumer electronics device equipped with two axial fans are presented to demonstrate the efficacy of the proposed scheme in practice. Note that the present article is a revised and expanded version of one originally presented at INTER-NOISE 2005³.

2 EFFECT OF AN ENCLOSURE

2.1 Source-Type Conversion

An axial fan operating in free-space exhibits a dipole-like radiation characteristic when the source can be considered to be compact. In most practical cases, however, a certain boundary condition is imposed on the sound field around the fan by the structure supporting it, thus resulting in a sound radiation pattern other than that of an ideal dipole. When an axial fan is mounted to an enclosure, as considered in the present work, the sound radiation pattern becomes close to that of a monopole since the interaction between the sound fields on both sides of a fan is, at least partially, prevented. Since a monopole is a more efficient radiator than a dipole, the conversion of the source type from a dipole to a monopole in this way results in an increase in the radiated sound power.

A simulation of this effect was performed by using a boundary element model to demonstrate the latter physical phenomenon: the simulation model is shown in Fig. 1. A dipole source was created by using two, closely-spaced monopoles (2 cm separation) that were operated out-of-phase with each other by 180 degrees;

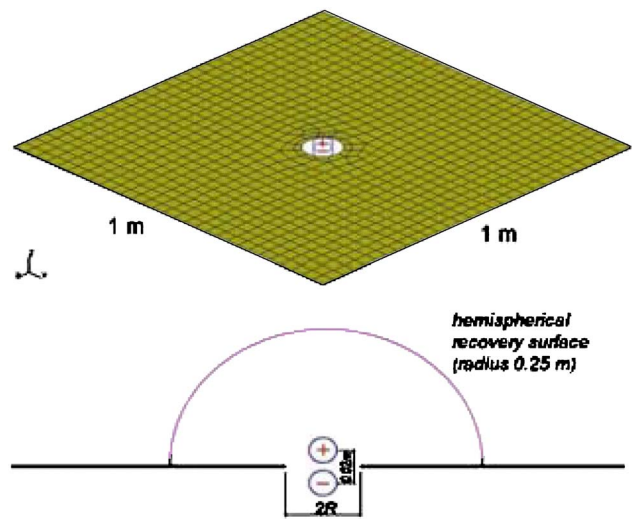


Fig. 1—A boundary element model for simulating the effect of a baffle on sound radiation.

a square (1-by-1 m), rigid surface having a circular hole at its center with a radius R was created to simulate a baffle; and the source was positioned at the center of the hole. The thickness of the rigid surface was assumed to be negligible. The radiated sound powers with respect to various sizes of the hole were calculated over a hemispherical recovery surface defined by a radius of 25 cm on one side of the surface.

In Fig. 2(a), the radiated sound powers of the dipole-baffle model when various hole sizes were used are shown with respect to the Helmholtz number, kL (where k and L are the acoustic wave number and the edge-length of the rigid baffle, respectively), along with a comparison to that of a dipole operating in free-space. It can be seen that the presence of the rigid baffle causes an increase in the radiated sound power, and that the radiated sound power progressively increases as the hole became smaller. Based on the latter observation, it can be concluded that the sound power radiated by a dipole can be substantially increased by the presence of a baffle, and that the degree of enhancement is related to the extent of the cancellation resulting from the interaction between the sound fields on the two sides of the dipole. In contrast, when the case of two monopoles operating in-phase was considered, no significant effect of the rigid baffle was observed since the pressure and normal particle velocity conditions that apply in the plane of the rigid surface are the same whether it is present or not (see Fig. 2(b)). Thus, it is suggested here that a fan's sound power can be substantially altered simply by mounting the fan to the exterior surface of an enclosure.

In this simulation, a finite baffle was used rather than an infinite baffle. Even in the former case, the effect of a baffle on the sound radiation from a dipole was

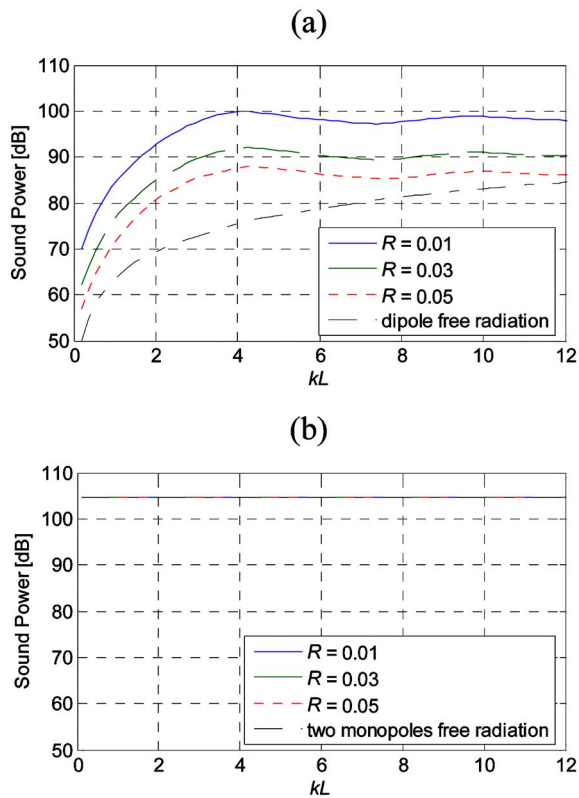


Fig. 2—A comparison of the radiated sound powers with respect to the size of the hole (R represents the radius of the hole in a baffle): (a) two out-of-phase monopoles (dipole); (b) two in-phase monopoles.

observed clearly, which implies that the latter effect may be observed in many practical cases when the interaction between the sound fields are only partially prevented (although its effect will be smaller than that in the infinite-baffle case). The effect of the finite baffle size can be clearly seen in Fig. 2(a): i.e., the frequency characteristics of the radiated sound power differ in the two regions divided by approximately $kL=4$. In the region below $kL=4$, the frequency characteristics of the radiated sound power in the presence of the baffle were similar to those of a free-field dipole, which leads to the conclusion that the rigid surface used in this simulation was not large enough compared to the acoustic wavelength in that frequency region to act like an infinite baffle; however, it nonetheless had an impact on the sound radiation characteristics. In contrast, in the region above $kL=4$, the frequency characteristics were almost uniform with respect to frequency, as expected for a monopole, which indicates that the baffle used in this simulation performed like an infinite baffle in that region, with the result that the source-type was converted from a dipole to a monopole. Thus, it can be expected that, in the absence of the hole, the radiated

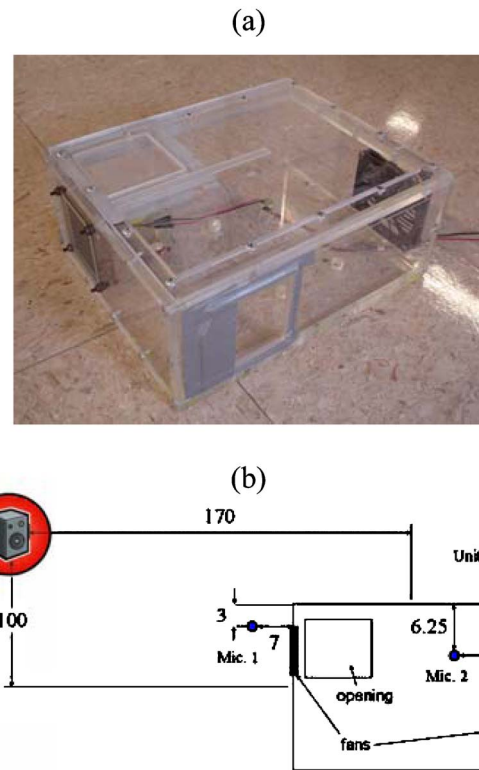


Fig. 3—Experimental setup for demonstrating the effect of acoustical cavity resonances: (a) Model enclosure; (b) measurement setup (top view).

sound power in the region above $kL=4$ is comparable to that observed in the two, in-phase monopole case.

Note that the effect just discussed may also impact the sound power estimated by using an INCE plenum, in which the fan under test is typically mounted on a baffle that may be relatively small compared to a wavelength at low frequencies⁴. The effect described in this subsection has subsequently been investigated elsewhere⁵.

2.2 Coupling with Interior Acoustical Resonances

The second effect of an enclosure is related to the interaction between the source and the interior acoustical resonances of the enclosure. The latter effect is demonstrated here experimentally by using a model enclosure. The size of the enclosure's internal cavity was 30 cm $\times$ 22.5 cm $\times$ 10 cm, and two fans were mounted to grilled ports located on opposite sides of the enclosure. A 7-by-7 cm, rectangular opening was created on the top of the enclosure (see Fig. 3(a)), and the opening was covered with a thin sheet of mylar, which is acoustically transparent, to prevent air flow through it. Further comments regarding the use of mylar will be made in Sec. 3.1.

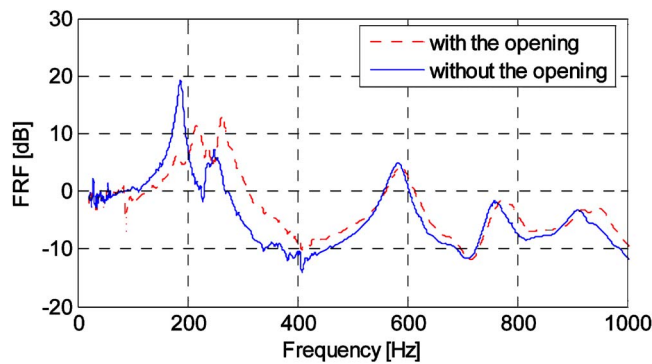


Fig. 4—A comparison of the frequency response functions of the model enclosure with and without the additional opening.

In Fig. 3(b), the measurement setup is shown. Random noise was generated by using a loudspeaker located outside the enclosure, and the frequency response function at Mic. 2 positioned inside the enclosure with respect to Mic. 1 positioned outside the enclosure was measured while the fans were not operating in order to identify the effect of interior acoustical resonances alone.

In Fig. 4, the frequency response functions obtained with and without the top opening are shown. When the opening was closed, the dominant peak appeared at 190 Hz. When the opening was introduced, a change in the frequency response characteristics resulted, especially at the dominant peak in the low frequency region: i.e., with the additional opening, the dominant peak was shifted to 262 Hz while the peak level of the frequency response function was decreased. Thus, it is suggested that the source level amplification caused by the coupling between the enclosure's resonances and the fan's operating speed can be mitigated by introducing an additional opening since it results in both a shift and increased damping of the dominant peak in the enclosure frequency response function.

The peaks observed in the frequency response function correspond to the resonances of the acoustical cavity. In this test-enclosure case, the lowest possible

resonance frequency resulting from standing wave behavior within the enclosure was 567 Hz (some deviation can be expected due to the effect of the two grilled ports at the fan locations), but the dominant peak was observed at a frequency much lower than that frequency. It was found that the dominant peak frequency, in fact, corresponded to the Helmholtz resonance frequency of the enclosure. The resonance frequency of a Helmholtz resonator with a single opening can be calculated by⁶

$$f_{res} = \frac{c}{2\pi} \sqrt{\frac{S}{L'V}} \quad (1)$$

where c is the speed of sound, S represents the area of the opening into the resonator, L' represents the effective length of the neck, and V represents the volume of the cavity. The Helmholtz resonance frequency of an enclosure having multiple openings, as considered here, can be obtained as in the single-opening case by using the following relation⁷:

$$\frac{S}{L'} = \sum_i \frac{S_i}{L'_i} \quad (2)$$

where S_i and L'_i represent the area and effective length of each opening, respectively. The values of the parameters used in this calculation are given in Table 1. The neck length of the additional opening corresponded to the panel thickness (1.3 cm), and the neck lengths of the grilled ports at the fan locations were chosen to be 3.3 cm to allow for the thickness of the fan housing (2 cm). The effective neck length of each opening was obtained by adding the end correction to the neck length. The end correction for arbitrarily-shaped, flanged openings of the Helmholtz resonator can be approximated by adopting the equivalent radius of the opening (i.e.)⁸, and thus the effective neck length was calculated as

Table 1—The values of the parameters used in the calculation of the Helmholtz resonance frequencies of the test enclosure.

Speed of sound (c)		343 m/s
Volume (V)		0.068 m ³
Panel thickness (t_p)		0.013 m
Fan housing thickness (t_h)		0.02 m
Grilled port $i=1,2$	Area (S_i)	$(0.072)^2 2/3$ (grille effect) = 0.0035 m ²
	Effective length (L'_i)	$(t_p + t_h) + 2(0.82) \sqrt{S_i/\pi} = 0.0874$ m
Additional opening $i=3$	Area (S_i)	$(0.07)^2 = 0.0049$ m ²
	Effective length (L'_i)	$t_p + 2(0.82) \sqrt{S_i/\pi} = 0.0778$ m

$$L'_i = L_i + 2(0.82) \sqrt{\frac{S_i}{\pi}} \quad (3)$$

where $\sqrt{S_i/\pi}$ represents the equivalent radius of the opening. The effect of the grille, which covered a third of the port area at the fan location, was approximated by assuming an opening whose area was two-thirds of the area of the port. The Helmholtz resonance frequencies of the enclosure calculated by using those parameters were, respectively, 186.9 and 250.5 Hz in the absence and presence of the additional opening: i.e., the additional aperture causes the Helmholtz resonance frequency to shift upwards. This predicted effect agreed well with the experimental results. The other evidence supporting the idea that the low frequency peak is a Helmholtz resonance is the fact that only the frequency of the first peak was changed significantly by the additional opening: i.e., while the Helmholtz resonance frequency is sensitive to the additional opening, its impact on the acoustic modes caused by the standing wave motion (e.g., the feature near 600 Hz) is relatively small.

The second measurement was performed while the fans were operating at 1036 rpm and 2100 rpm, and the radiated sound powers with and without the opening are compared in Fig. 5. Since the fans had eleven blades, the blade passing frequencies corresponding to the two rotational speeds were, respectively, 190 Hz and 385 Hz. The sound power measurements were performed by using a hemispherical array of ten microphones according to the ISO standard (ISO 3744)⁹. When the fans were operating at 1036 rpm, a significant reduction of the radiated sound power was achieved by introducing the opening (Fig. 5(a)): i.e., the level at the blade passing frequency was reduced by approximately 17 dB. A part of that reduction may have been a result of source-type conversion, but it is judged that in this case the main effect was due to the shift of the enclosure's resonance away from the blade passing frequency and the increased damping of that resonance, as shown in Fig. 4. When the fans were operating at 2100 rpm, the reduction in the sound power around 190 Hz resulted for the same reason as in the previous case, but the sound powers at the blade passing frequency (i.e., 385 Hz) remained nearly the same. At the blade passing frequency, amplification of the source level could be expected by the introduction of the opening according to the frequency response characteristics of the enclosure, but the effect of the latter amplification seemed to be cancelled out by the effect of source-type conversion.

When the results at the two fan operating speeds obtained without the additional opening are compared, the effect of the enclosure's cavity resonance is evident:

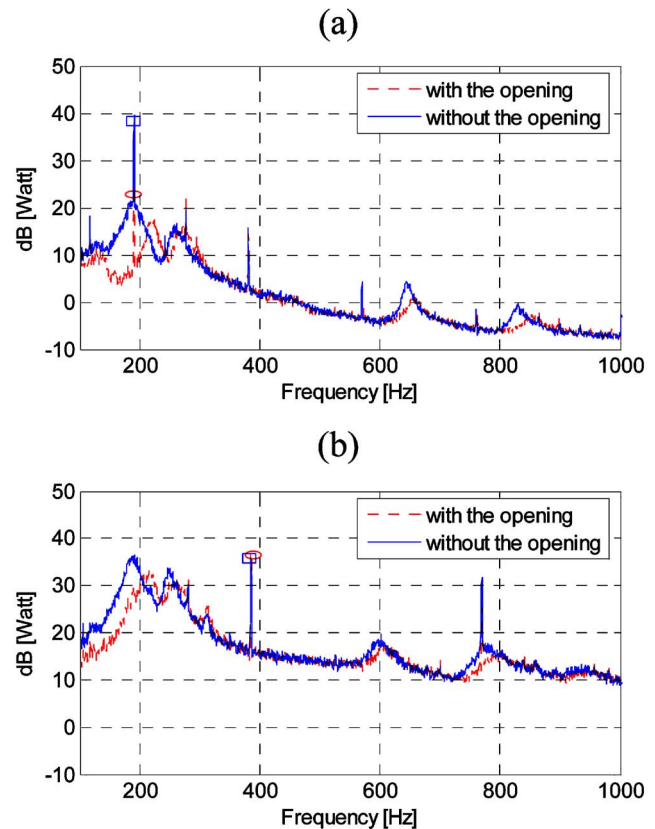


Fig. 5—A comparison of the radiated sound power of the model enclosure with and without the additional opening: (a) when the fans were running at 1036 rpm (BPF = 190 Hz); (b) when the fans were running at 2100 rpm (BPF = 385 Hz).

i.e., a larger radiated sound power level resulted when the fans were operating at 1036 rpm rather than 2100 rpm, in spite of the fact that the source strength itself becomes larger as the fan speed increases. Thus, it can be concluded that the radiated sound power is a result of the combined effect of the frequency characteristics of the source and the resonance frequencies of the enclosure, and that the size of the additional opening must be optimized in consideration of the latter effect.

3 APPLICATION OF THE STRUCTURAL MODIFICATION SCHEME TO A CONSUMER ELECTRONICS DEVICE

In the present section, experimental results obtained by the application of the structural modification scheme proposed here to a consumer electronics device are presented. A series of measurements were made with various types of apertures to illustrate some of the concepts discussed above. To identify the change in fan noise radiation patterns that occurred when apertures

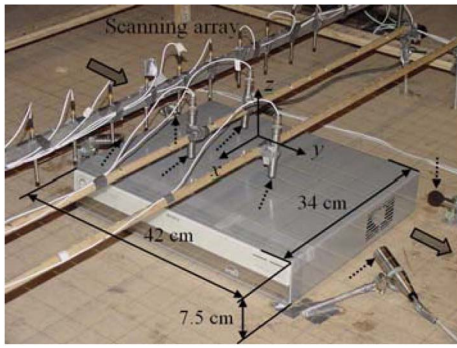


Fig. 6—The test consumer electronics device and the NAH measurement setup (the thick, gray arrows depict the direction of air flow through the two fans, and the arrows depicted with a dashed line indicate the reference microphone locations).

were created in the enclosure, the resulting sound fields were visualized by using near-field acoustical holography (NAH)¹⁰. Also, the radiated sound powers were calculated by integrating the acoustic intensity fields, which were reconstructed by using NAH, over a surface, and their levels for various enclosure modifications were compared with the baseline level to quantify the effect of apertures.

3.1 Experimental Setup for Near-Field Acoustical Holography

In Fig. 6, the consumer electronics device used in the present study and the measurement setup are shown. The electronics enclosure had a dimension of 42 cm × 34 cm × 7.5 cm, and two, 6-cm-diameter fans having eleven blades were installed on opposite sides of the unit. The origin of the coordinate system was set to the central location between the two fans on the top surface of the enclosure. The fans mounted on the left side (at $x=0$ cm and $y=-21$ cm) and on the right side (at $x=0$ cm and $y=21$ cm) were intake and exhaust fans, respectively; the thick, gray arrows shown in Fig. 6 depict the direction of air flow through the two fans.

In the present work, planar NAH based on the use of discrete Fourier transform (DFT) was implemented in conjunction with a scan-based, cross-spectral procedure¹¹. The unit was placed on a rigid surface, and seven reference microphones were distributed close to possible source locations (the arrows depicted with dashed lines indicate the reference microphone locations). Two reference microphones were placed near each fan, and three reference microphones were placed over the top of the enclosure. The sound field on the planar hologram surface located 2 cm above the top of the enclosure was measured by using a scan-based procedure at 11-by-16, uniform grid points with incre-

ments of 8 cm in both x - and y -directions (which allowed the sound field to be visualized in a frequency range up to 2.1 kHz): i.e., a linear array comprising eleven, equally-spaced (8 cm) field microphones was traversed in the y -direction in sixteen steps with an increment of 8 cm. From the latter measurement, the cross-spectral matrices between the reference microphones, and between the reference and field microphones, were estimated with a frequency resolution of 2 Hz. The composite sound field obtained in the latter way was decomposed into a set of mutually incoherent partial sound fields by using the virtual coherence method that allows a sound field to be decomposed based on the use of the basis functions provided by singular value decomposition (SVD)^{12,13}. Each partial sound field on the hologram surface was then extended into an out-of-measurement region by using a procedure referred to as patch NAH in order to suppress the windowing effect resulting from the truncation of a sound field at the edges of the finite measurement aperture^{14,15}. Finally, the extended partial sound fields were projected onto a reconstruction surface individually by using NAH, and the total sound field was obtained by adding the significant partial sound fields together on an energy basis. The number of significant partial fields required to construct the total sound field can be identified as the number of fields that causes the sum of the virtual coherence functions to approach a unity value over the entire hologram surface¹³.

The enclosure was modified in three ways: first, the whole top of the enclosure was replaced by an acoustically-transparent mylar sheet, thus allowing the sound fields on both sides of the fans to interact with each other freely while the air flow was maintained within the enclosure (see Fig. 7(a)). The next two modifications were chosen by considering their applicability to real products. Grilled ports and perforated panels were added to the top of the enclosure near the fan locations as shown in Figs. 7(b) and 7(c). In both cases, a mylar sheet was also applied to the openings to prevent air from flowing out through them. It is noted that mylar was used since the change in flow caused by the openings would have had an impact both on cooling performance, which is important for electronics devices of this type, and on the acoustic performance. The introduction of additional openings without mylar coverings may or may not improve those two aspects at the same time, but it is also usually required that these devices be protected from dust or water spills, for example. In any events, the same current was provided to the two fans so that they operated at the same nominal rotational speed, but a small difference in the two fans' speeds in each case was observed presumably due to slight changes of the flow conditions experi-

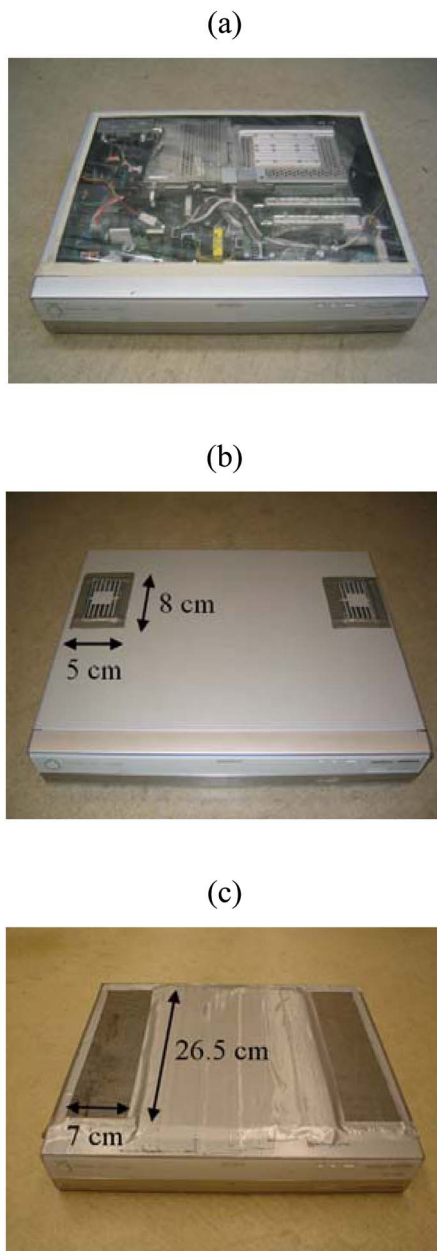


Fig. 7—Modifications of the test electronic device enclosure: (a) mylar top case; (b) grilled port and mylar case; (c) perforated panel and mylar case.

enced by the fans. Through the measurements, the two fans rotated around 1790 rpm within a 1.5-percent variation: that effect is negligible in the various comparisons shown later.

3.2 Results

3.2.1 Singular values of the reference cross-spectral matrix

In Fig. 8, the singular values of the cross-spectral matrix constructed between the seven reference signals

are shown. An examination of the latter singular values allows the number of incoherent sources contributing to the total sound field to be estimated¹⁶, and the singular values can be interpreted as the source strengths of the corresponding partial sound fields in most cases. From the singular value plots, it was concluded that the total sound fields in all cases resulted mainly from two incoherent sources in the frequency region that is of interest in the present work: thus the two partial sound fields corresponding to the first two singular values were required to construct the total sound field. The latter result is consistent with the fact that the test electronics device had two running fans. Since the second singular value was much smaller than the first singular value, it could also be expected that the sound fields generated by the two fans would exhibit a considerable difference in their levels.

3.2.2 Partial sound fields

The conclusions drawn from the examination of the singular values can be confirmed by using the virtual coherence function. Since the virtual coherence function represents the contribution of each partial sound field to the total sound field over all the measurement locations, the use of the virtual coherence function can provide more reliable conclusions compared to the use of the singular values by themselves. In Fig. 9, an example of the use of the virtual coherence function is shown. It can be seen that the sum of the virtual coherence functions calculated by using the first two partial sound fields at 328 Hz (i.e., the blade passing frequency) in the original-enclosure case was close to unity over the entire hologram surface (see Fig. 9(b)); the latter observation led to the conclusion that the first two partial sound fields (i.e., the sound fields corresponding to the two incoherent sources) were required to construct the total sound field. When the contributions of the first two partial sound fields to the total sound field were compared, it was found that the contribution of the first partial field (Fig. 9(a)) was much larger than that of the second partial field: i.e., the virtual coherence of the first partial field was close to unity everywhere except in the region near the exhaust fan, and the addition of the second partial field resulted in only a slight improvement. Although not presented here, similar results were found in the other three cases considered.

In Figs. 10 and 11, the spatial distributions of the first two partial sound fields in the various cases are shown at the blade passing frequency. In those figures, real parts of the partial fields are plotted, and the horizontal and vertical planes represent the source surface (i.e., the top of the enclosure) and an orthogonal plane joining the two fans. As suggested by an

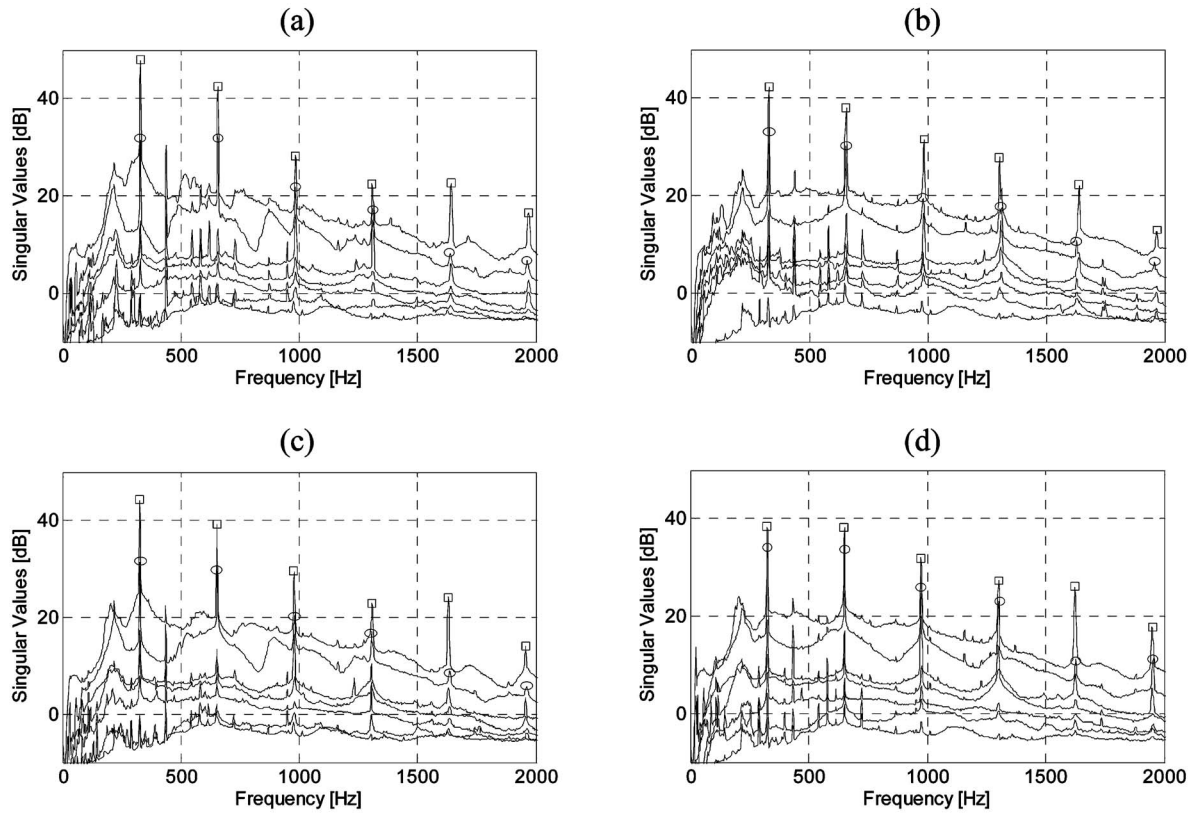


Fig. 8—The singular values of the reference cross-spectral matrix constructed from the test electronics device measurements: (a) original enclosure case; (b) mylar top case; (c) grilled port and mylar case; (d) perforated panel and mylar case. The level of the two largest singular values at the blade passing frequency harmonics are denoted by □ and ○.

examination of the virtual coherence function, it can be seen that the amplitudes of the first partial sound fields were much larger than those of the second partial sound fields in all cases. Based on the results obtained when the opening was large (i.e., the mylar-top and perforated-panel cases), the relative strengths of the sound fields radiated from the intake and exhaust fans can be identified since the sound fields radiated from the two fans were well-separated in those cases (i.e., the first and second partial sound fields consisted essen-

tially of the dipole-like sound fields radiated by the intake and exhaust fan, respectively). It was found that the intake fan was the more significant noise source for the device used in this study. The latter result could be anticipated since the air flowing into the intake fan was disturbed due to the grille located immediately upstream of the fan inlet. Therefore, optimizing the shape of a grille located in front of an intake fan for the

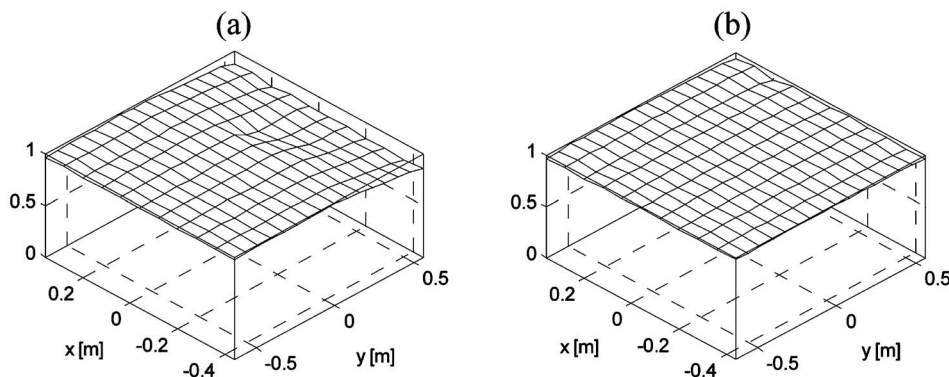


Fig. 9—The sum of the virtual coherence functions in the original enclosure case at the blade passing frequency (328 Hz): (a) when the first partial field was used; (b) when the first and second partial fields were used.

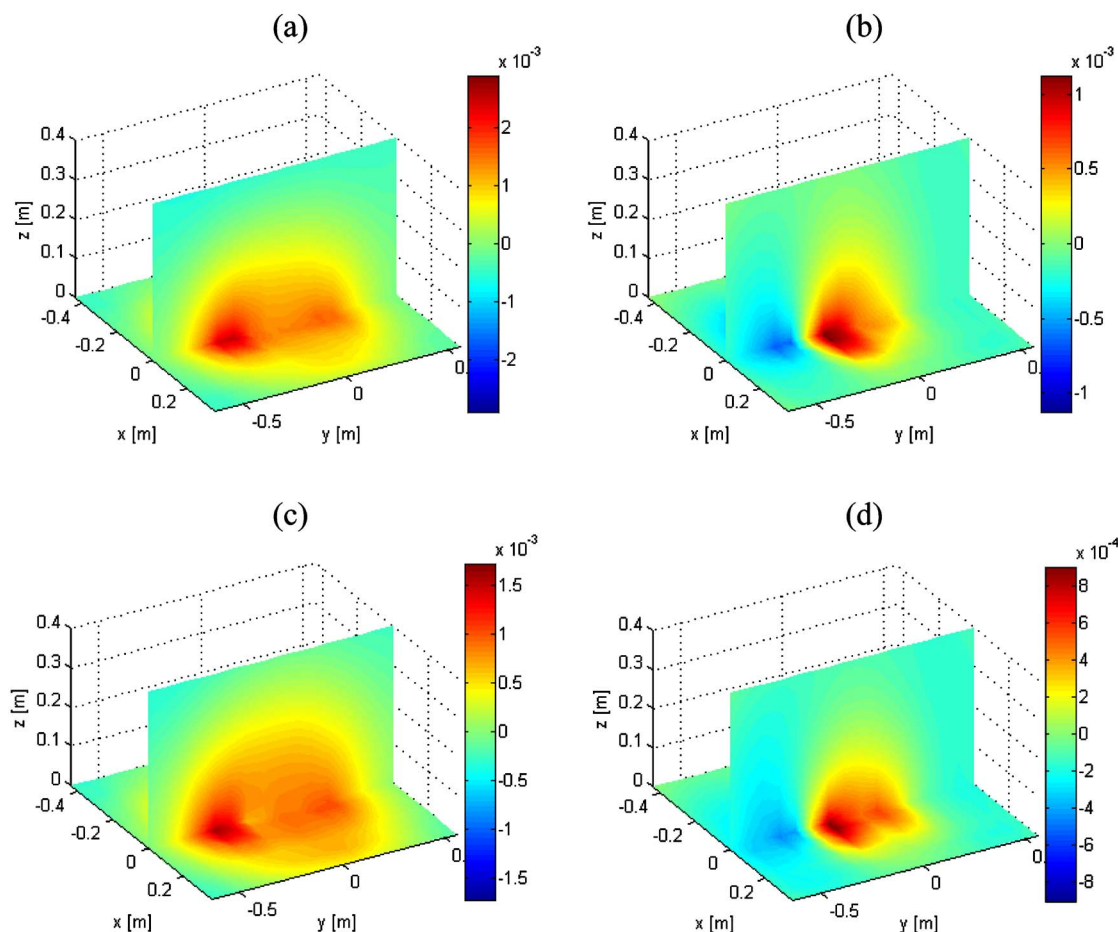


Fig. 10—The first partial sound field of the test electronics device at the blade passing frequency (real parts are plotted): (a) original enclosure case (at 328 Hz); (b) mylar top case (at 328 Hz); (c) grilled port and mylar case (at 326 Hz); (d) perforated panel and mylar case (at 324 Hz).

purpose of controlling the inlet flow condition could be an effective noise control solution for a product of this type.

When the original enclosure was used, it can be seen that the sound fields exhibited a monopole-like radiation pattern. In the grilled-port case, the sound fields remained monopole-like overall since the size of the opening was small compared to the other two modification cases. In those two cases, it can be seen that the partial sound fields comprised sound fields radiating simultaneously from both fan locations, which provides evidence of the role of the acoustical cavity (i.e., the sound field of each fan was amplified within the acoustical cavity and radiated from both fan locations simultaneously, as is visible in Figs. 10(a), 10(c), 11(a), and 11(c)).

3.2.3 Total sound field

The total sound field was obtained by adding quadratically the first two partial sound fields. In Fig. 12, the acoustic intensities of the total sound fields in

the various cases are shown at the blade passing frequency along with the amplitudes of the total sound pressures. In those figures, the results are shown on the plane defined by $x=0$, i.e., in the plane joining the fans at $y=\pm 21$ cm, and the arrows that depict the acoustic intensity were scaled to make it possible to compare the relative strengths of the acoustic intensities in the various cases. The differences between the sound fields can be identified more clearly in those figures: i.e., a larger opening resulted in a clearer conversion to dipole-like radiation and a greater nearfield effect. That effect is particularly clear in Figs. 12(b) and 12(d), where an approximate null in the radiation field is visible at $y=-21$ cm, as would be expected for a dipole oriented in the x -direction. In both the original-enclosure and grilled-port cases (i.e., Figs. 12(a) and 12(c), respectively), the radiation from the two fans were, by contrast, clearly monopole-like. Nonetheless, even though the overall radiation pattern remained

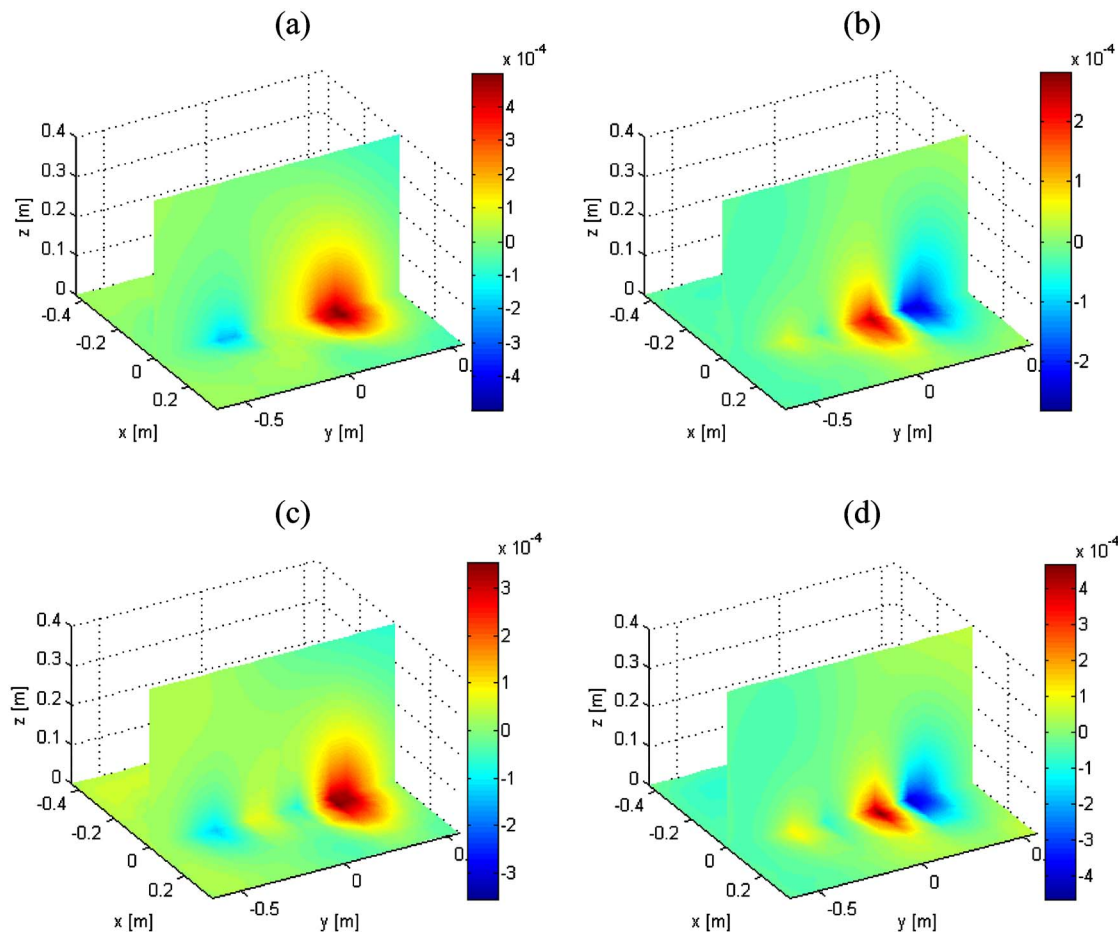


Fig. 11—The second partial sound field of the test electronics device at the blade passing frequency (real parts are plotted): (a) original enclosure case (at 328 Hz); (b) mylar top case (at 328 Hz); (c) grilled port and mylar case (at 326 Hz); (d) perforated panel and mylar case (at 324 Hz).

monopole-like in the grilled-port case, a significant reduction in radiated sound level at the blade passing frequency was achieved.

In Fig. 13, the radiated sound powers at the blade passing frequency and its harmonics are compared to quantify the effect of the enclosure modifications. The sound power was calculated by including both the first and second partial sound fields in the frequency band between the half-power points of the tones. At the fundamental and twice the blade passing frequencies, it can be seen that a larger opening resulted in a correspondingly greater reduction of the sound power, while at the same time causing a slight increase in the sound power at the higher harmonics. The sum of the total sound powers of the first five tones considered in Fig. 13 was reduced from 34.5 dB to 24 dB in the mylar-top case, to 30.3 dB in the grilled port case, and to 25.7 dB in the perforated-panel case.

4 CONCLUSIONS

The effect of an enclosure on the sound radiation from fans mounted to it is expected to appear in various

ways depending on the precise details of each situation. In the low frequency region where the free-field sound radiation pattern of a fan can be described by a point dipole, an enclosure renders the radiation pattern monopole-like by preventing the interaction between the sound fields on both sides of a fan, thus enhancing the fan's radiation efficiency; the sound level at the source can then be further amplified by coupling with the enclosure's interior acoustical resonances (the Helmholtz resonance, in particular). In the present work, it was suggested that the latter enhancement of radiation efficiency can be suppressed by introducing a properly designed acoustical path to an enclosure. At high frequencies, in contrast, the enhancement of the radiated sound power caused by the effect of an enclosure is not likely to be significant, not only since the source-type conversion is not clear compared to that at low frequencies, but also since, except for very small devices, the lowest interior resonance frequency typically appears below 1 kHz. In the latter case, the sound muffling effect of an enclosure may be larger

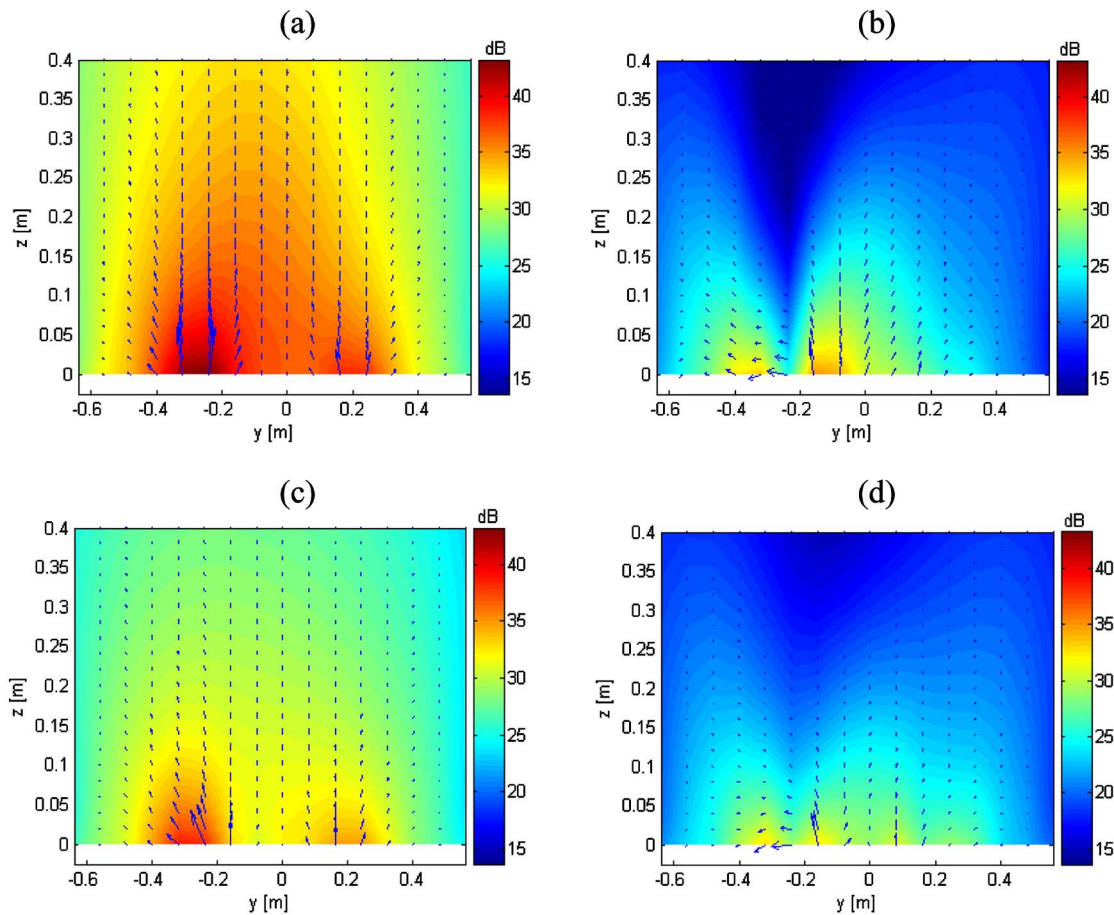


Fig. 12—The acoustic intensity along with the amplitude of the total sound field of the test electronics device at the blade passing frequency on the plane defined by $x=0$.: (a) original enclosure case (at 328 Hz); (b) mylar top case (at 328 Hz); (c) grilled port and mylar case (at 326 Hz); (d) perforated panel and mylar case (at 324 Hz).

than the amplification, so that the opening may cause an increase in the radiated sound power or poor sound quality by radiating internally created flow noise

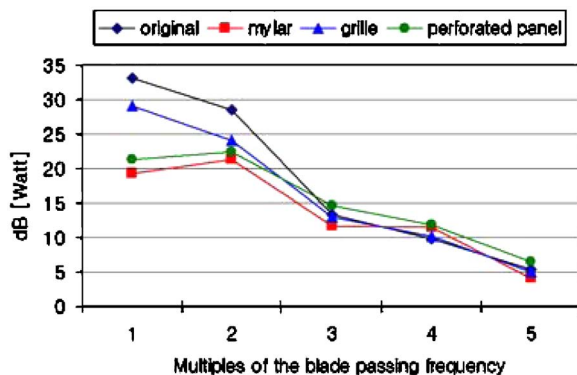


Fig. 13—A comparison of the radiated sound powers of the test electronics device with various enclosure configurations at the blade passing frequency and its harmonics.

through it, for example. As a result, the modification scheme described in the present study may sometimes not be suitable. Therefore, in a practical implementation, the characteristics of a situation encountered should be identified first, and the size and location of the enclosure opening should then be chosen accordingly to achieve an optimal result. Also, the use of other types of material (e.g., porous materials) in place of a mylar sheet may be considered to help suppress the radiation of the high frequency noise through an opening while still preventing significant air flow through it.

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